



Alessandro Conti

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GENERAL INFORMATION

Birth Day 2002/01/11
Nationality Italy
Languages Italian, English

SUMMARY

Master's degree in Computer Science and Engineering from the Politecnico di Milano, with a solid background in *cybersecurity, operating systems, and software engineering*. Passionate about low-level programming, system architecture, and the development of secure and scalable software. Motivated to apply the academic rigour and practical experience gained while writing my thesis to solve complex technical challenges in dynamic engineering contexts.

PROJECTS

CryptAttackTester

[Repository](#)

This project is my master's thesis. The aim was to integrate parallelization for the existing ISD algorithms into the [CryptAttackTester](#) framework

Constant_Time-Parallel_Shuffling

[Repository](#)

Shuffling arrays in constant time is a common problem in modern cryptography. This project involves analysing the technique proposed by Daniel J. Bernstein in Python, and implementing it in C employing parallelization.

The project includes porting the DJB code from Python to C, in both sequential and parallel versions. In addition to simple transcription, the project also analyses the computational complexities of the algorithms, providing a basis for optimisation and benchmarking.

Progetto_API_2023

[Repository](#)

This project manages a motorway with single-distance stations, each equipped with electric rental vehicles with variable range. A journey is a sequence of stages with no possibility of turning back, renting a vehicle at each stage. The goal is to find the route with the minimum number of stages between two stations; in the event of a tie, preference is given to the stages closest to the start of the motorway.

Progetto_Ingegneria_del_Software_2023

[Repository](#)

This project implements an online version of the board game 'My Shelfie'.

Progetto_Reti_Logiche_2023

[Repository](#)

This project implements a hardware module in VHDL that acts as an interface between a memory and four output channels. The module receives serial instructions relating to a memory address and the desired output channel.

The system receives input data via a single serial bit indicating the memory address from which to retrieve the data and the corresponding output channel. The system outputs consist of four parallel channels that transmit the entire memory word.

Fault-Tolerant_Data_Flow

[Repository](#)

This project implements a distributed data flow platform for processing large amounts (big data) of key-value pairs, where keys and values are integers.

The platform includes a coordinator and multiple workers running on multiple nodes of a distributed system. The coordinator accepts data flow programmes specified as an arbitrarily long sequence of the above operators.

Code_Kata_Battle

[Repository](#)

Code Kata Battle is a platform that helps students improve their software development skills by training with peers on code data.

Sorting_Network

[Repository](#)

This project implements several sorting networks.

The sorting networks implemented are: odd-even transposition sort, odd-even mergesort, bitonic sort, LS3 sort, 4-way mergesort, rotatesort, 3n sort of Schnorr and Shamir, 2D odd-even transposition sort and shearsort.

EDUCATION

2023 - 2026/07/22	Master of Science (Computer Science and Engineering - Ingegneria Informatica) at Politecnico di Milano (110/110)
2020 - 2023/09/27	Bachelor's Degree (Ingegneria Informatica) at Politecnico di Milano
2015 - 2020	High School (Liceo Scientifico opzione Scienze Applicate) at I.I.S. James Clerk Maxwell, Milan

SKILLS

Programming Language	C, C++, Java, Python, Bash, JavaScript
Frontend Languages	HTML, CSS
Query Languages	SQL
Operating System	Linux, Windows, macOS
Documentation	Latex, Markdown
Version Control	Git, GitHub
Database Management System	MySQL
Software	WireShark, TeamViewer, Microsoft Office